

House Price Prediction Using Machine Learning: A Study on the Impact of Spatial and Environmental Features

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Abstract—This study investigates the problem of predicting house prices using machine learning techniques. The dataset used in this study contains various housing features, including structural characteristics, environmental factors, and spatial information such as city and waterfront. A complete data modelling pipeline was applied, including data preprocessing, feature engineering, and model training. Several regression models, including Linear Regression, Random Forest, and Gradient Boosting, were implemented and compared. The results show that ensemble models outperform linear models due to their ability to capture non-linear relationships. In addition, spatial and environmental features such as city and waterfront were found to have a significant impact on house prices. These findings highlight the importance of incorporating both structural and spatial features in house price prediction.

Index Terms—house price prediction, machine learning, regression, spatial features, environmental factors

I. INTRODUCTION

House price prediction is a key task in the real estate industry. Real estate plays an important role in the global economy and affects many stakeholders, including buyers, sellers, investors and policymakers [1]. Moreover, in reality, house prices are influenced by a wide range of factors, including complex geographical factors, local regulatory policies and neighborhood characteristics. Hence, accurate property valuation becomes particularly challenging yet essential for supporting informed decision-making, ensuring economic transparency and maintaining economic stability. Specifically, as a highly differentiated product, the value of real estate is shaped by both internal factors (such as building quality and layout) and external environment attributes (such as geographical location and employment accessibility) [2]. Traditional approaches, which highly rely on expert judgement or simplified statistical methods, such as averaging out decades of data or the neighborhood data, often struggle to capture the intricate interactions among these factors. Combined with limited market information and low transaction frequency, such methods are prone to subjective bias and inconsistent valuations [1].

To address these limitations, machine learning has been increasingly adopted for house price prediction. Unlike traditional statistical approaches, these methods are capable of capturing complex and non-linear relationships between house features, thereby achieving significantly higher predictive accuracy. A wide range of algorithms have been explored in this domain, including linear regression, decision trees, random forest, and gradient boosting models. However, despite these advances, no single model can consistently achieve the best performance across all scenarios. Each model type has inherent trade-offs: linear models offer interpretability but struggle with nonlinear patterns [3]; tree-based models capture complex interactions but risk overfitting [4], [5]; and neural networks provide flexibility at the cost of requiring careful and delicate architecture design and hyperparameter tuning. To overcome these individual limitations, ensemble learning methods have been introduced into house price prediction. Among them, stacking is particularly promising, as it uses a meta-learner to integrate predictions from heterogeneous base models through a multilayer architecture. By combining diverse models that capture different aspects of the data, stacking can correct individual prediction biases and improve overall accuracy.

In order to overcome the limitations of a single prediction model, the integrated learning method has been widely introduced into the field of house price prediction. Among them, Stacking uses a meta-learner to integrate predictive output from heterogeneous learners (such as decision trees, random forests, gradient boosting, and neural networks) by building a multi-layer learning architecture. This mechanism can capture and correct the prediction bias of different algorithms and significantly improve the accuracy and robustness of the general prediction through the synergistic complementary effect [6], [7].

Despite its potential, the systematic application of stacking ensembles to house prediction remains limited [8]. Most existing studies focus on individual models or simpler ensemble techniques such as Bagging and Boosting [5], with relatively few investigating stacking in this context. Moreover, the studies that do apply stacking have reported mixed findings:

while some improvements in accuracy have been observed, the gains are often marginal and come at a high computational cost, raising questions about the methods' general applicability in the real market [9]. More critically, existing stacking-based studies tend to treat the ensemble as a black box, with insufficient analysis of how each base learner contributes to the overall predictions. This limits both the theoretical understanding and practical adoption of the methods in housing valuation [9]. These gaps motivate the present study, which aims to evaluate whether a stacking ensemble can consistently outperform individual models and to analyse the specific role of each base model within the ensemble.

To investigate these research gaps, this study used the King County housing dataset obtained from Kaggle, which contains 4600 house transactions recorded between May 2014 and May 2015. The dataset included 18 features describing each property, covering structural attributes (e.g., bedrooms, bathrooms, and living area), as well as spatial and environmental factors (e.g., waterfront and view). The target variable is the house sale price. Hence, this dataset is particularly suited for evaluating stacking ensembles as various feature types allow different base models to leverage different aspects of the data.

Specifically, this study addresses the following research questions:

- How do individual machine learning models compare in predicting house prices?
- Can a stack ensemble that combines the base models outperform the best-performing individual models?
- Which base models contribute most to the stacking ensemble's performance?

II. LITERATURE REVIEW

With the rapid development of data-driven technology, machine learning (ML) methods have emerged as a cornerstone of contemporary house price prediction. The non-parametric methods excel at capturing the intricate, non-linear dependencies inherent in real estate features, thereby substantially enhancing predictive fidelity compared to traditional frameworks [10]. To date, researchers have validated a diverse suite of algorithms, ranging from foundational linear regression [1] to sophisticated tree-based models such as Decision Trees [2], Random Forests, and Gradient Boosting [11]. When processing large-scale, high-dimensional housing data, these ML approaches exhibit superior robustness and accuracy to traditional parametric models (such as hedonic regression), primarily due to their inherent capacity to model complex spatial and structural interactions.

Despite these advancements, the “No Free Lunch” theorem persists in property valuation: no individual model consistently achieves optimal performance across all market scenarios. Each architecture presents distinct trade-offs. Linear models offer unparalleled interpretability and computational efficiency but lack the flexibility to resolve complex non-linearities [3]. Conversely, tree-based ensembles effectively capture non-linear feature interactions but are susceptible to overfitting

and typically offer lower interpretability [4], [5]. While Multi-layer Perceptrons (MLP) can learn highly complex mappings, their efficacy is contingent upon delicate network architecture design and rigorous hyperparameter tuning [3]. Furthermore, Support Vector Regression (SVR) can process non-linearities in higher-dimensional spaces via kernel functions, yet it remains sensitive to the selection of the penalty parameter C and the ϵ -insensitive zone, while suffering from high computational costs as sample sizes increase [12]. All these performance variances highlight the critical need for a more integrated modeling approach.

To transcend the limitations of individual learners, ensemble learning methods have been extensively adopted to stabilize predictions and minimize variance [13]. Techniques such as Bagging and Boosting, exemplified by Random Forests and Gradient Boosting Machines, have established high benchmarks in housing valuation due to their capacity to aggregate multiple decision trees and navigate complex feature interactions [11], [14]. Building upon this, Stacked Generalization (Stacking) represents a more sophisticated architectural evolution. By employing a meta-learner to integrate predictive outputs from heterogeneous base models, this multi-layer framework facilitates the integration of complementary algorithmic strengths. This mechanism effectively corrects individual prediction biases, thereby enhancing both the accuracy and structural robustness of the final valuation model [15]. However, despite these methodological advancements, critical gaps remain in the systematic application of stacking ensembles to real estate forecasting. First, the existing literature frequently lacks rigorous, controlled comparisons between models under consistent experimental settings, which impedes the formulation of generalizable conclusions [8]. Second, the application of stacking in this specific domain remains relatively underexplored [16], with most studies treating the ensemble as a “black box” models without sufficient interpretability. Specifically, there is a distinct lack of granular analysis regarding how individual base learners contribute to the overall ensemble performance. To address these deficiencies, this study systematically evaluates the consistent superiority of stacking ensembles over individual baselines and employs an ablation study to dissect the functional role of each base learner within the integrated framework.

III. METHODOLOGY

This study adopts a structured end-to-end data modelling pipeline for house price prediction. The pipeline includes exploratory data analysis, data preprocessing, feature engineering, feature selection, baseline model construction, stacking-based ensemble learning, and model evaluation. Each stage is designed to improve predictive performance, reduce noise, and minimise the risk of data leakage.

The analysis begins with exploratory data analysis (EDA) to gain an overall understanding of the dataset structure and the relationships between variables. The dataset is first loaded from `data.csv` and examined through summary statistics, data types, and sample records, providing an initial overview

of its characteristics. To assess data quality, missing values are analysed by computing both their counts and proportions across all features. As no significant missingness is observed, no imputation is required. Attention is then directed towards the target variable (*price*). Its distribution is visualised using a histogram with kernel density estimation, and skewness is calculated to quantify asymmetry. This step helps identify potential issues such as imbalance or extreme values. Building on this, the relationships between key predictors and house price are explored. Regression plots are used to examine continuous variables such as *sqft_living*, while boxplots are employed to analyse the influence of categorical variables, including *waterfront* and *view*. Finally, a correlation heatmap is constructed for numerical variables, allowing important predictors to be identified and potential multicollinearity to be detected. Together, these steps provide a comprehensive understanding of the dataset and inform subsequent modelling decisions.

To reduce the impact of extreme values, outliers are removed by excluding observations above the 99th percentile of *price*. This improves model stability and is particularly important for error-sensitive metrics such as RMSE.

Following data cleaning, feature engineering is conducted to enrich the representation of the dataset and better capture underlying patterns in house prices. The *date* variable is first converted into a datetime format, enabling the extraction of temporal attributes such as *year* and *month*. These temporal features are then used to derive additional variables, including *house_age*, *is_renovated*, and *years_since_renovation*, which reflect the lifecycle characteristics of properties. In addition to temporal transformations, several structural features are introduced to describe the internal composition of houses. For instance, *basement_ratio* captures the proportion of basement space relative to total living area, while *has_basement* provides a binary indication of its presence. Similarly, *bath_bed_ratio* represents the balance between bathrooms and bedrooms, offering a more informative measure than absolute counts. A small constant is incorporated in ratio calculations to ensure numerical stability and avoid division-by-zero issues. Finally, to account for potential seasonal effects in the housing market, the month variable is mapped into four categories, including winter, spring, summer, and autumn, forming a new *season* feature. This allows the model to capture temporal variations in demand that may influence property prices.

After feature engineering, irrelevant and redundant variables are removed. Raw location and identifier variables such as *street*, *city*, and *statezip* are excluded at this stage, as they are either high-cardinality or replaced by engineered representations. The categorical *season* feature is encoded using one-hot encoding, and boolean variables are converted into numeric form to ensure compatibility with machine learning models. The dataset is then split into training and testing sets using an 80/20 ratio with a fixed random seed.

To preserve spatial information without introducing high dimensionality, target encoding is applied to *statezip* and *city*. The mean house price for each category is computed using

only the training data and then mapped to both training and test sets. Unseen categories in the test set are assigned the global mean price. This approach captures location-specific effects while preventing data leakage.

Feature selection is performed using a hybrid strategy combining statistical filtering and domain knowledge. Features with an absolute correlation greater than 0.05 with the target variable are retained. In addition, several domain-relevant features are manually preserved to ensure that important predictors are not discarded. To address multicollinearity, pairwise correlations among selected features are analysed, and highly correlated variables (correlation above 0.85) are removed using an upper-triangular filtering approach. However, key variables such as *sqft_living*, *bathrooms*, and the target-encoded features are protected to maintain model interpretability.

Before modelling, feature scaling is applied using standardisation. The scaler is fitted on the training data and applied to both training and test sets to avoid leakage. This step is particularly important for models sensitive to feature magnitude, such as Ridge regression and neural networks.

A different set of baseline models is then constructed to capture different learning behaviours. Ridge regression is used as a regularised linear baseline. Random Forest and Extra Trees represent bagging-based ensemble methods capable of modelling nonlinear interactions. Gradient Boosting and XGBoost are included as boosting-based approaches that iteratively reduce residual errors. In addition, an MLP regressor is used to capture complex nonlinear patterns through neural network representation learning. All models are trained on the scaled features and evaluated using R^2 , RMSE, and MAE to provide complementary performance perspectives.

To further enhance performance, a stacking ensemble framework is implemented. Five-fold cross-validation is used to generate out-of-fold (OOF) predictions for each base model. For each fold, the model is trained on a subset of the data and used to predict the held-out fold, producing unbiased meta-features for training. Test predictions are generated in each fold and averaged to ensure stability.

Three stacking configurations are explored. The first uses only OOF predictions as inputs to a Ridge meta-learner. The second extends this by introducing a passthrough mechanism, where original scaled features are concatenated with OOF predictions, allowing the meta-learner to utilise both sources of information. The third replaces the Ridge meta-learner with LassoCV, enabling automatic regularisation and feature selection within the ensemble. This design improves flexibility and allows the model to adaptively select informative predictors.

Model performance is evaluated on the held-out test set using R^2 , RMSE, and MAE, and results are compared across all baseline and ensemble models. To analyse the contribution of each base model, an ablation study is conducted by removing one model at a time from the stacking ensemble and measuring the resulting performance drop. This provides insight into the relative importance of each component. In addition, feature importance is assessed using a Random Forest model, and diagnostic plots including predicted-versus-actual,

residual scatter, and residual distribution are used to evaluate model behaviour and identify potential biases.

Overall, the proposed methodology integrates systematic preprocessing, domain-informed feature engineering, leakage-aware encoding, hybrid feature selection, diverse baseline models, and stacking-based ensemble learning. These design choices collectively improve model robustness, generalisation, and interpretability.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section presents the experimental process and results of the proposed modelling pipeline. It begins with exploratory data analysis and feature engineering insights to understand data characteristics and guide feature design. Subsequently, multiple regression models are implemented and compared, including both baseline methods and stacking-based ensembles, with performance evaluated using standard metrics.

A. Exploratory Analysis and Feature Engineering Insights

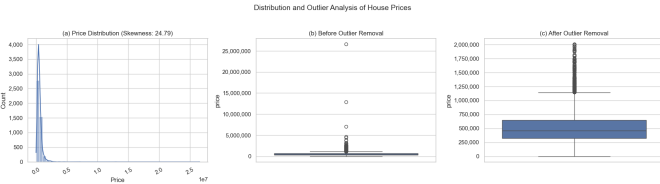


Fig. 1: Distribution and outlier analysis of house prices. (a) The original price distribution is highly right-skewed, indicating the presence of extreme values. (b) Before outlier removal, a large number of extreme high-price observations are visible. (c) After removing the top 1% of prices, the distribution becomes more concentrated and stable.

The experimental process begins with an exploratory analysis of the dataset to understand the distribution of the target variable and the relationships between key features. As shown in Fig. 1, the distribution of house prices is highly right-skewed, with a dense concentration of lower-valued properties and a long tail of extreme high prices. This is further confirmed by the boxplot, where numerous outliers extend far beyond the upper quartile. Such extreme values can disproportionately affect model training and lead to unstable predictions, particularly for error-sensitive metrics such as RMSE.

To address this issue, a percentile-based filtering strategy is applied by removing the top 1% of price values. After preprocessing, the distribution becomes more compact with reduced variance, indicating improved data stability for subsequent modelling. Following this, the relationships between key predictors and house price are examined in Fig. 2. A clear positive relationship is observed between *sqft_living* and price, although the dispersion suggests non-linear effects. In addition, properties with waterfront access and higher view ratings consistently achieve higher prices, indicating that both structural and location-related features are important determinants of house value. These observations motivate the inclusion of such variables in the modelling stage. To

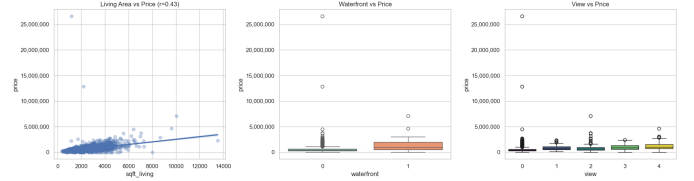


Fig. 2: Relationships between key features and house price. (a) Living area shows a positive correlation with price. (b) Properties with waterfront access have significantly higher prices. (c) Price increases with higher view ratings, indicating a monotonic trend.

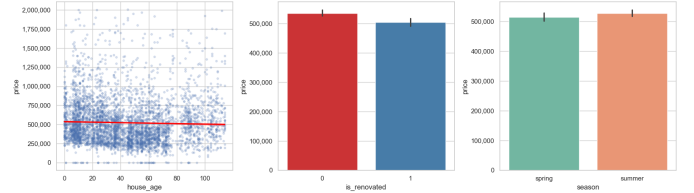


Fig. 3: Impact of engineered features on house prices. (a) House age shows a weak negative relationship with price. (b) Renovated properties exhibit slightly higher average prices. (c) Seasonal variation indicates marginal differences across time periods.

further enhance the representational capacity of the dataset, several engineered features are introduced and evaluated in Fig. 3. Features such as *house_age*, *is_renovated*, and seasonal indicators show relatively weak individual relationships with price. However, these variables are retained as they may provide complementary information or capture interaction effects when used in more complex models. Spatial variation is further analysed in Fig. 4, where substantial differences in average house prices are observed across cities. The ranked distribution reveals a long-tailed pattern, indicating that a small number of locations contribute disproportionately to high property values. This confirms that location is a key determinant of price. Given the high cardinality of spatial variables, direct one-hot encoding would be inefficient; therefore, target encoding is adopted to preserve location information while controlling feature dimensionality. To quantify feature importance, the absolute correlation between each feature and the target variable is presented in Fig. 5. The results show that *sqft_living* and the target-encoded spatial features (*zip_te* and *city_te*) have the strongest associations with price, followed by variables such as *bathrooms*, *bedrooms*, and *view*. In contrast, several engineered features exhibit weaker correlations but are retained due to their potential contribution in non-linear or ensemble models. Finally, the effectiveness of feature selection is evaluated by comparing the correlation structures before and after refinement, as shown in Fig. 6. The initial feature set exhibits strong multicollinearity, particularly among size-related variables (e.g., *sqft_living* and *sqft_above*), with correlations reaching up to 0.88. After applying correlation-

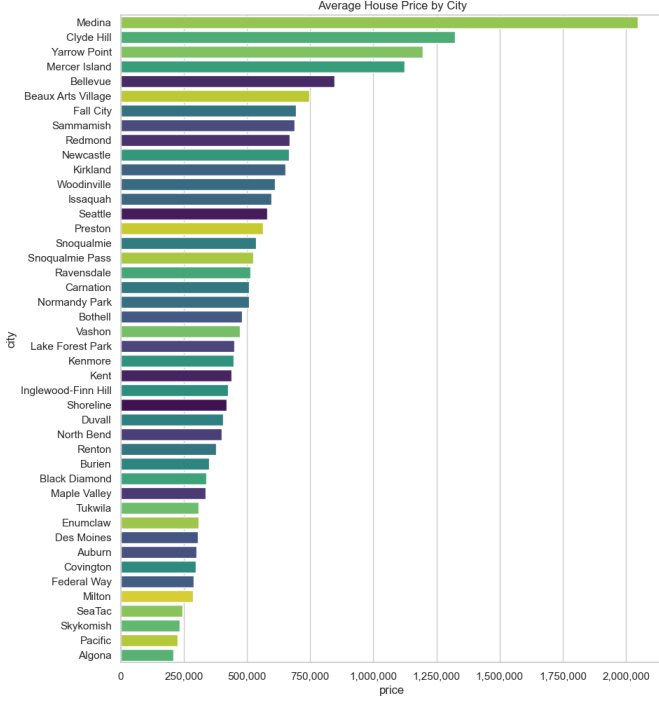


Fig. 4: Average house price across different cities. Significant variation in average prices is observed, indicating strong location-dependent effects.

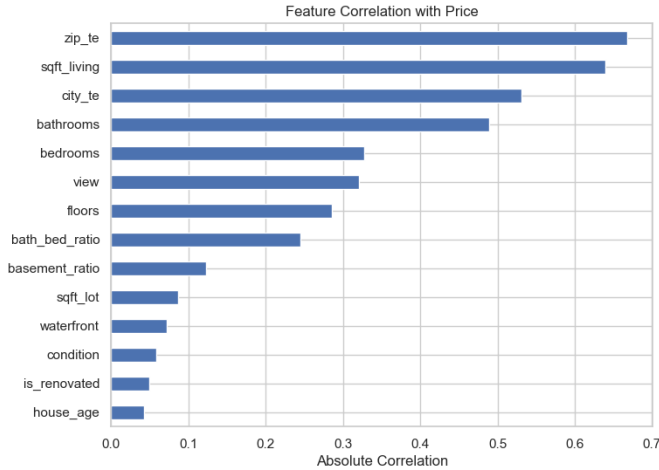


Fig. 5: Absolute correlation between features and house price. Strong correlations are observed for *sqft_living*, *zip_te*, and *city_te*, while several engineered features exhibit moderate or weak relationships.

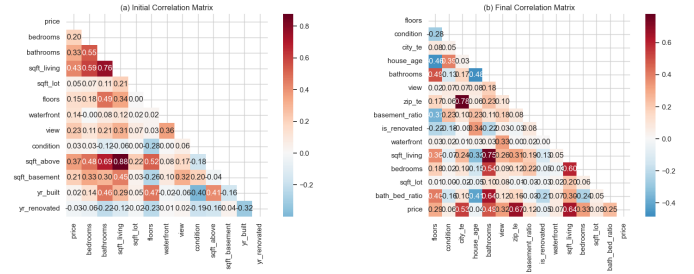


Fig. 6: Comparison of correlation structures before and after feature selection. (a) The initial correlation matrix shows strong multicollinearity among several features (e.g., *sqft_living* and *sqft_above*). (b) After feature selection, redundant variables are removed, resulting in a more balanced and less correlated feature space.

based filtering, redundant variables are removed, resulting in a more balanced and less correlated feature space. Importantly, key relationships with the target variable are preserved. This reduction in multicollinearity improves model stability and interpretability, particularly for linear models.

B. Model Performance and Comparative Evaluation

Following the exploratory analysis and feature engineering stage, the refined feature set is used to train and evaluate multiple regression models. These include a linear model (Ridge), several tree-based ensemble methods (Random Forest, Extra Trees, Gradient Boosting, and XGBoost), a neural network model (MLP), and stacking-based ensembles. This section presents a comparative evaluation of these approaches, followed by an analysis of prediction behaviour and the impact of preprocessing.

TABLE I: Performance comparison of models

Model	R^2	RMSE	MAE
Stacking V1	0.7467	146316	88025
Stacking V3	0.7462	146440	87915
Stacking V2	0.7439	147113	88716
XGBoost	0.7417	147741	88430
GBR	0.7366	149207	89691
Random Forest	0.7295	151188	88370
Extra Trees	0.7053	157818	93898
MLP	0.7018	158732	100606
Ridge	0.7003	159141	100720

The overall performance results are summarised in Table I and visualised in Fig. 7. Among the individual models, XGBoost achieves the highest R^2 score (0.7417), followed by Gradient Boosting and Random Forest. This confirms the effectiveness of tree-based ensemble methods in modelling nonlinear relationships and complex feature interactions.

Stacking models further improve performance beyond all individual approaches. In particular, Stacking V1 achieves the best overall performance ($R^2 = 0.7467$), while Stacking V3 performs comparably. Stacking V2 shows slightly lower performance, likely due to differences in the meta-learning configuration. Although the improvement over the best individual

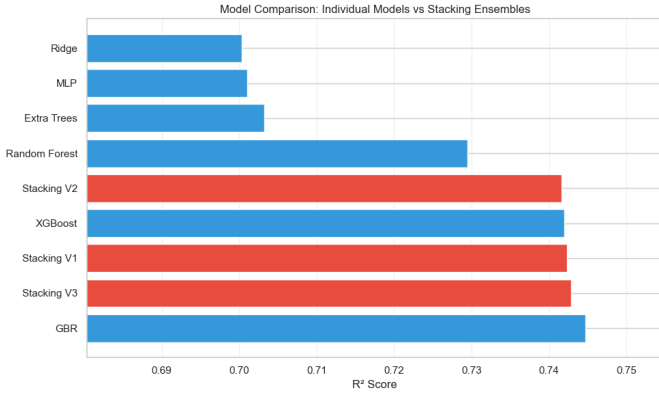


Fig. 7: Comparison of model performance in terms of R^2 . Stacking-based ensembles consistently outperform individual models.

model is moderate, the consistent gains across all metrics demonstrate that stacking effectively combines complementary model behaviours. This suggests that the ensemble is better able to capture both linear and nonlinear patterns than any single model.

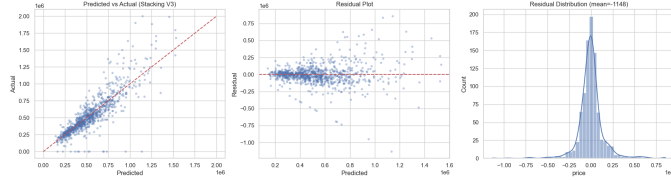


Fig. 8: Prediction performance and residual analysis of a representative stacking model (Stacking V3).

To further assess model behaviour, Fig. 8 presents diagnostic plots for a representative stacking model. The predicted-versus-actual plot shows that most observations lie close to the diagonal line, indicating strong predictive accuracy.

The residual plot does not exhibit a clear systematic pattern, suggesting that the model captures the main structure of the data. However, the dispersion of residuals increases for higher predicted values, indicating mild heteroscedasticity. The residual distribution is centred around zero, confirming that the model does not exhibit significant bias. Overall, the results indicate that the stacking model provides reliable predictions, with slightly reduced accuracy at extreme price levels.

The importance of preprocessing is highlighted in Fig. 9. A substantial improvement in performance is observed after applying data cleaning and feature engineering, with R^2 increasing from near zero to above 0.73 for both Random Forest and XGBoost. This indicates that model performance is primarily constrained by data quality rather than model complexity, and that appropriate preprocessing is essential for effective learning.

To analyse the contribution of individual base learners, an ablation study is conducted as shown in Fig. 10. Each model is

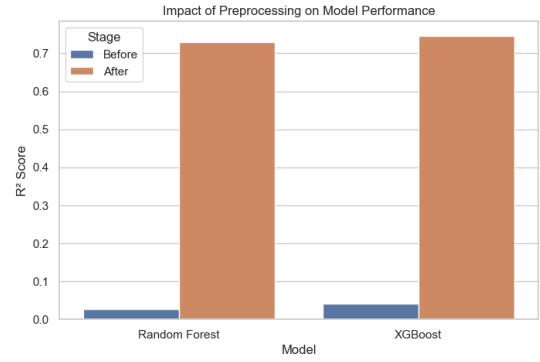


Fig. 9: Impact of preprocessing on model performance.

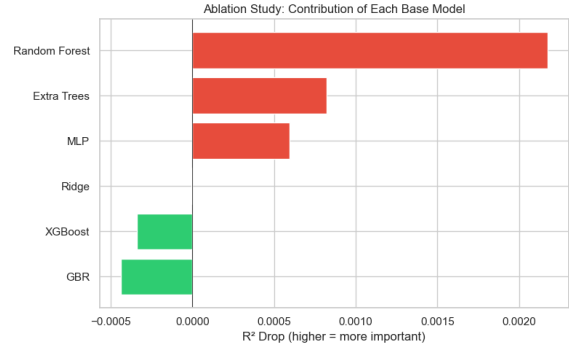


Fig. 10: Ablation study showing the contribution of each base model to the stacking ensemble. Positive R^2 drop indicates performance degradation when the model is removed.

removed from the ensemble in turn, and the resulting change in performance is measured.

The results indicate that removing XGBoost leads to the largest drop in performance ($\Delta R^2 = 0.0021$), highlighting its dominant contribution within the ensemble. Random Forest and Extra Trees also contribute positively, although their impact is comparatively smaller.

In contrast, removing Ridge and Gradient Boosting results in a slight improvement in performance, suggesting that these models may introduce redundant or overlapping information when combined with stronger learners. The contribution of MLP is negligible, as its removal produces almost no change in performance.

Overall, the ablation analysis demonstrates that the effectiveness of the stacking model is primarily driven by a small subset of strong base learners, particularly XGBoost. This highlights the importance of carefully selecting complementary models rather than simply increasing ensemble size.

In summary, the experimental results demonstrate that model performance is strongly influenced by both data quality and model design. Effective preprocessing and feature engineering significantly improve predictive capability, while ensemble methods further enhance performance by integrating complementary information from multiple models. Among all approaches, stacking achieves the best results, with its

effectiveness largely driven by a small number of strong base learners, particularly XGBoost. These findings highlight the importance of combining robust data preparation with carefully selected models to achieve reliable and accurate house price prediction.

V. DISCUSSION

The experimental results provide several important insights into the effectiveness of different modelling approaches for house price prediction. Rather than focusing solely on performance metrics, this section interprets the observed patterns and examines the underlying reasons for model behaviour.

First, the strong performance of tree-based ensemble methods, particularly XGBoost and Gradient Boosting, highlights the importance of capturing nonlinear relationships and feature interactions in housing data. As observed in the exploratory analysis, key variables such as *sqft_living*, *view*, and location-based features exhibit complex and non-linear effects on price. Linear models such as Ridge regression, which assume additive and linear relationships, are therefore inherently limited in their ability to model such patterns. This explains their comparatively lower performance. These findings are consistent with prior studies in house price prediction, which report that boosting-based models often outperform linear approaches due to their flexibility in modelling complex data structures.

Second, the improvement achieved by stacking models suggests that different algorithms capture complementary aspects of the data. While tree-based models are effective at modelling nonlinearities, linear models can still contribute by stabilising predictions, and neural networks may capture subtle patterns not explicitly represented in tree splits. However, the relatively modest gain from stacking indicates that the strongest individual models already capture most of the underlying signal. This is further supported by the ablation analysis, where the removal of XGBoost leads to the largest performance drop, confirming its dominant role within the ensemble. In contrast, removing Ridge or Gradient Boosting slightly improves performance, suggesting redundancy or overlapping information among certain base learners.

Third, the diagnostic analysis reveals that although the stacking model achieves strong overall accuracy, its performance is not uniform across all price ranges. The presence of heteroscedasticity, with larger residual variance at higher price levels, indicates that expensive properties remain more difficult to predict. This can be attributed to the limited number of high-value samples and the greater variability in luxury housing characteristics, which are not fully captured by the available features. Similar challenges have been reported in previous research, where models tend to perform worse on extreme values due to data imbalance and increased noise.

Furthermore, the significant improvement observed after preprocessing demonstrates that data quality plays a more critical role than model complexity. The removal of outliers and the introduction of engineered features substantially improve model performance, transforming the dataset into a more learnable representation. This supports the widely

recognised principle in machine learning that effective feature engineering and data cleaning are essential prerequisites for high-performing models.

Finally, the results also highlight the importance of feature representation, particularly for spatial variables. The strong contribution of target-encoded location features confirms that geographic information is a key determinant of house prices. Compared to traditional encoding methods, target encoding provides a compact and informative representation, enabling models to effectively utilise high-cardinality categorical variables without introducing excessive dimensionality.

Overall, the findings suggest that successful house price prediction depends on three key factors: high-quality data preprocessing, appropriate feature representation, and the use of flexible models capable of capturing nonlinear relationships. While ensemble methods such as stacking offer additional improvements, their effectiveness relies on the inclusion of strong and diverse base learners rather than simply increasing model complexity.

VI. CONCLUSION AND FUTURE WORK

This study presents a structured end-to-end modelling pipeline for house price prediction, integrating data preprocessing, feature engineering, feature selection, and multiple regression models, including stacking-based ensembles. The experimental results show that model performance is strongly influenced by both data quality and model design.

Among the evaluated approaches, tree-based ensemble models demonstrate superior performance, confirming their ability to capture nonlinear relationships and feature interactions within the data. The stacking framework further improves predictive accuracy by combining complementary strengths of multiple models, although the overall gain is moderate. This indicates that while ensemble learning is effective, its performance largely depends on the inclusion of strong base learners rather than increasing model complexity alone.

The results also highlight the critical importance of preprocessing and feature representation. Techniques such as outlier removal, target encoding, and feature engineering significantly improve model performance, suggesting that well-prepared data is essential for effective learning. In particular, spatial features encoded through target encoding are shown to be key contributors, reinforcing the importance of location in house price prediction.

Despite these strengths, several limitations remain. First, the model exhibits reduced accuracy for high-value properties, as indicated by increased residual variance. This is likely due to data imbalance and the limited representation of extreme cases. Second, the current approach relies solely on structured tabular features and does not incorporate external data sources such as economic indicators or geographic information, which may further enhance predictive performance. Third, the stacking framework, while effective, shows diminishing returns, suggesting potential redundancy among certain base models.

Future work can address these limitations in several directions. Incorporating additional data sources, such as temporal

economic trends or spatial coordinates, may improve model generalisation. More advanced feature engineering techniques, including interaction features or learned embeddings, could further enhance representation. In addition, exploring alternative ensemble strategies, such as weighted stacking or model selection based on diversity, may lead to more efficient integration of base learners. Finally, specialised modelling approaches for high-value properties could be developed to better handle heteroscedasticity and improve prediction accuracy in extreme cases.

Overall, this study demonstrates that effective house price prediction requires a combination of high-quality data preprocessing, appropriate feature representation, and robust modelling techniques, with ensemble methods providing additional but controlled performance gains.

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